

NATURAL LANGUAGE PROCESSING

4. Large Language Models (LLMs)

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Lecture Outline

- LLM Overview
- Training and Fine Tuning
- Evaluation Benchmarks
- LLM Design Optimizations
- Training Infrastructure

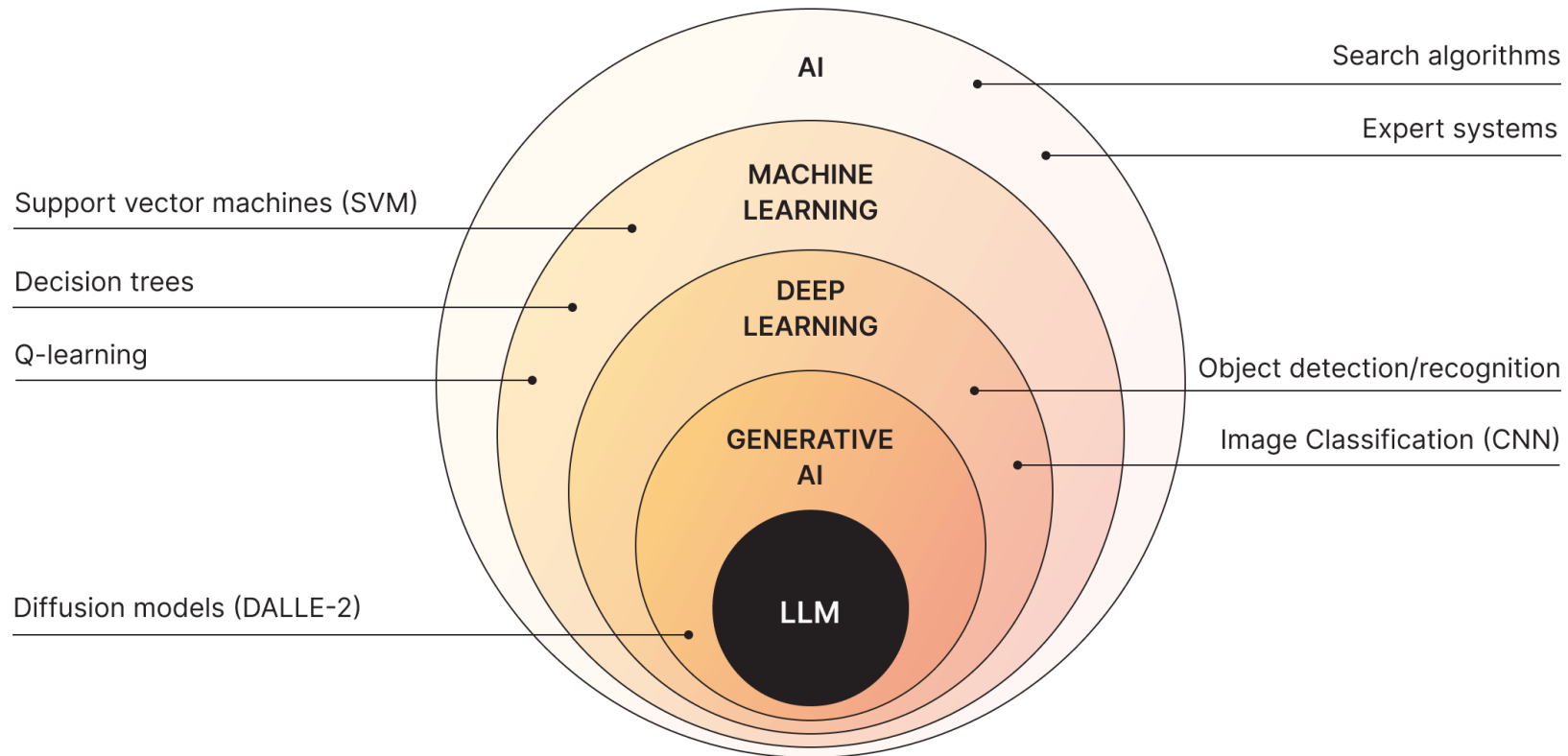


LLM Overview

Foundation Models

- Large-scale, pre-trained neural networks that serve as the base for fine-tuning specific tasks in AI applications.
- Benefits:
 - Reduced need for training models from scratch
 - Improved efficiency and flexibility across a wide range of tasks
- Challenges:
 - High computational cost during training
 - Ethical concerns, including bias and misuse

Large Language Models (LLM)



Source: memgraph.com

LLM: Applications

- Ad Generation & Brand Building
- Accent Modulation
- Voice Augmentation
- Audio, Video, Photos Editing
- Build Apps from Text Interface
- Business Intelligence
- Code Generation
- Contact Center
- Sales Coaching
- Sales Tools
- Content Repurposing
- Search Engine
- Content Marketplaces
- Documentation Writing
- Music generation
- Synthetic Voices & Dubbing
- Training Data
- Translation
- Voice of the Customer Analysis
- Semantic Search
- Knowledge Bases

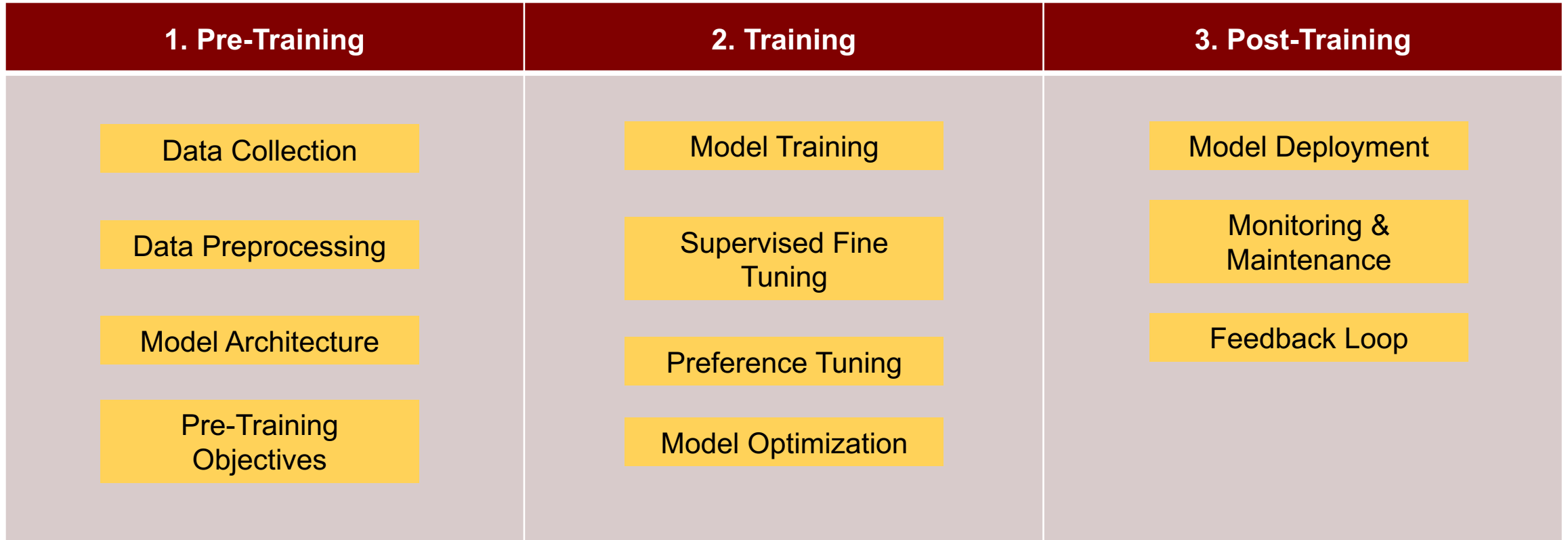
LLM: Challenges

- As models get larger, full fine-tuning becomes expensive
- Prompt Engineering is not an exact science and is model dependent
- Privacy and security concerns with sharing proprietary data with LLMs
- Storing and deploying fine-tuned models independently for each downstream task becomes very expensive

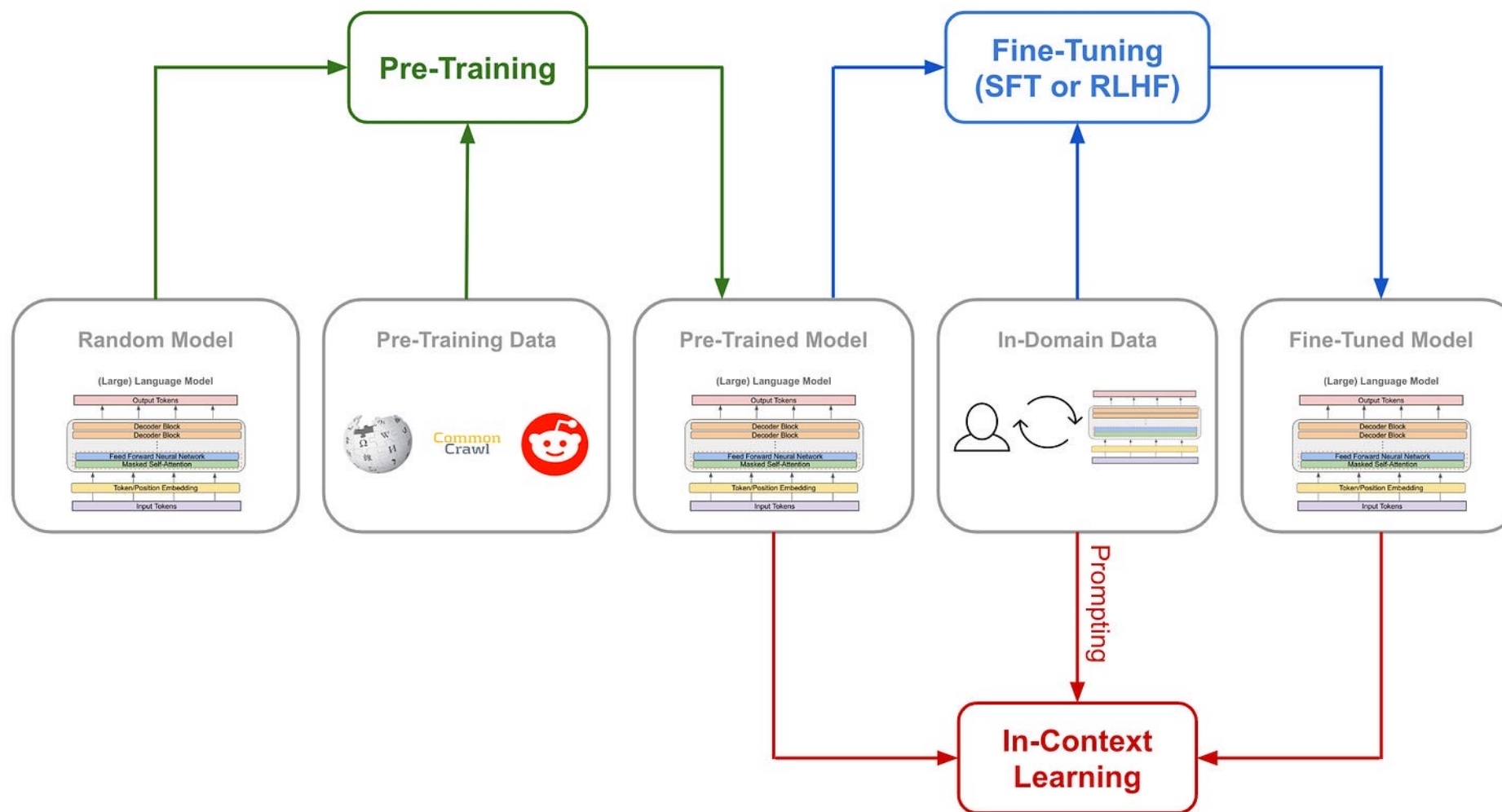


LLM Training and Fine Tuning

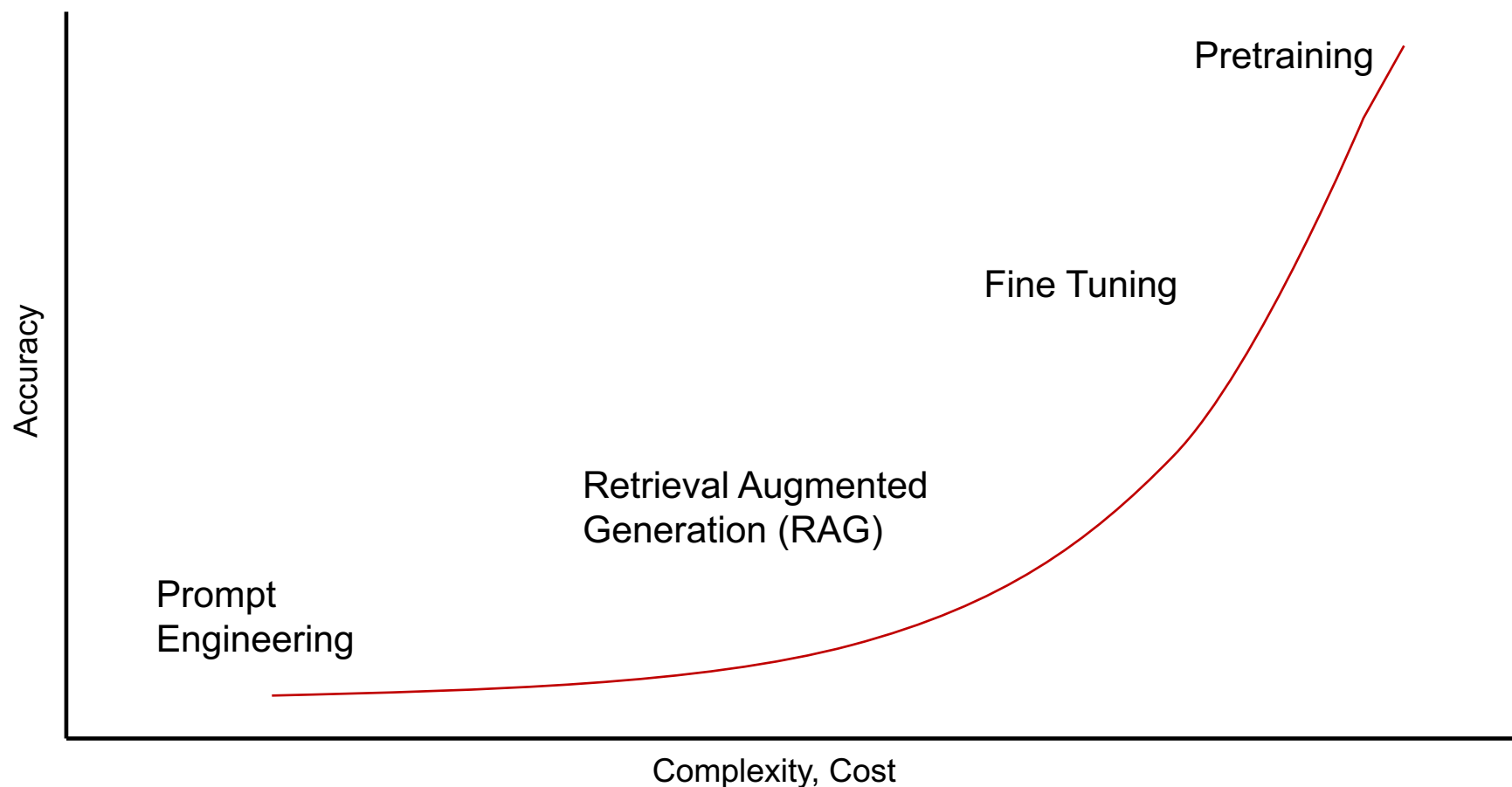
LLM Development Process



Pre-Training vs Fine-Tuning



LLM Accuracy vs Complexity



Supervised Fine-Tuning (SFT)

- A pre-trained model is fine-tuned on a smaller, task-specific labeled dataset using supervised learning to enhance its performance on a specific task.
- Domain Specialization
 - E.g., legal, medical, or financial text.
- Task Adaptation
 - E.g., text classification, named entity recognition, etc.
- Applications
 - Creative Writing, News Generation, Medical/Legal Texts
- Limitations
 - Static Behavior and imitation
 - No human preferences

SFT: Steps

1. Pretrained Model

- Start with a pre-trained model trained on large-scale, general data

2. Task-Specific Dataset

- Fine-tune the model on a smaller, domain-specific dataset with labeled examples

3. Supervised Learning

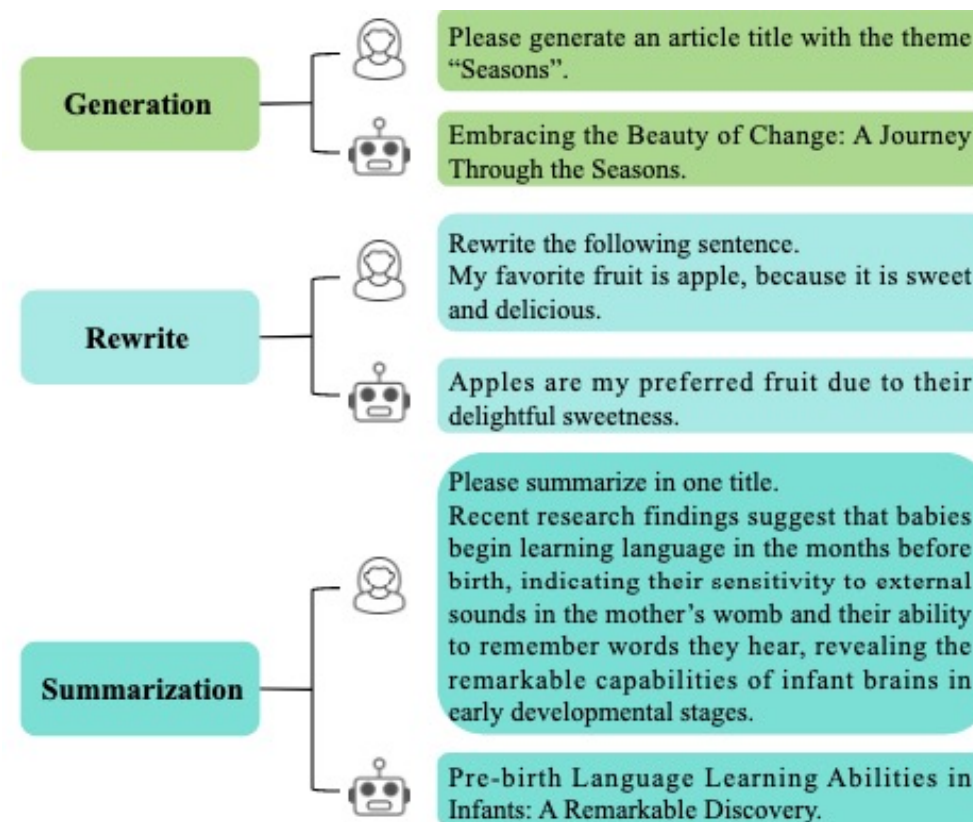
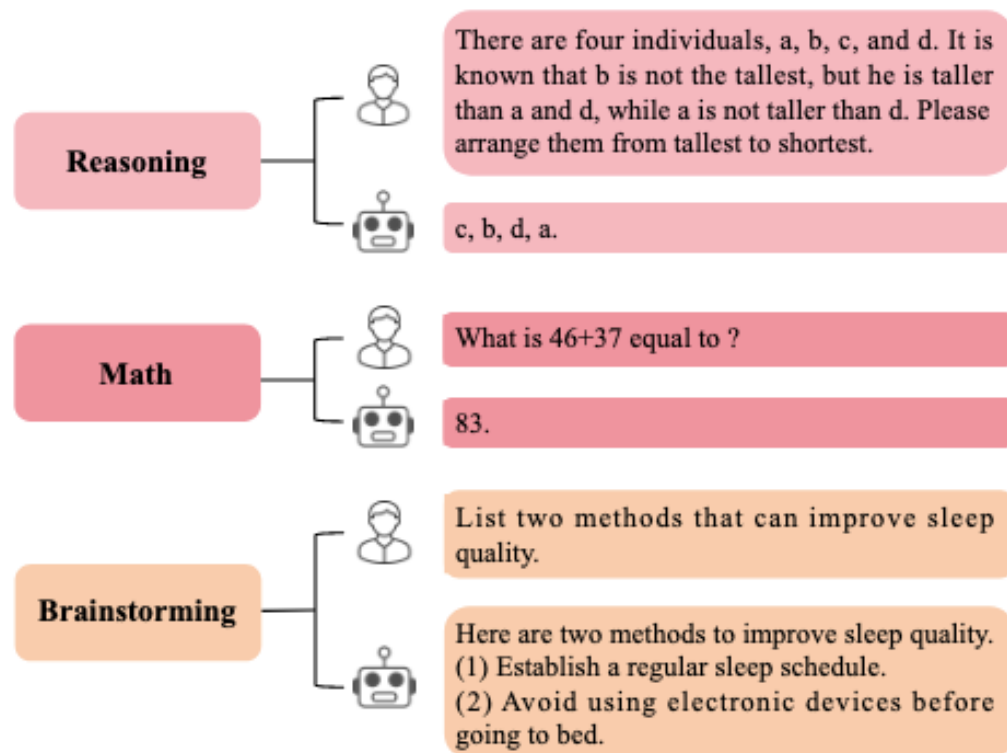
- Update model weights using Supervised Learning based on the labeled data.

4. Performance Evaluation

- Evaluate the model on task-specific metrics (accuracy, F1-score, etc.).

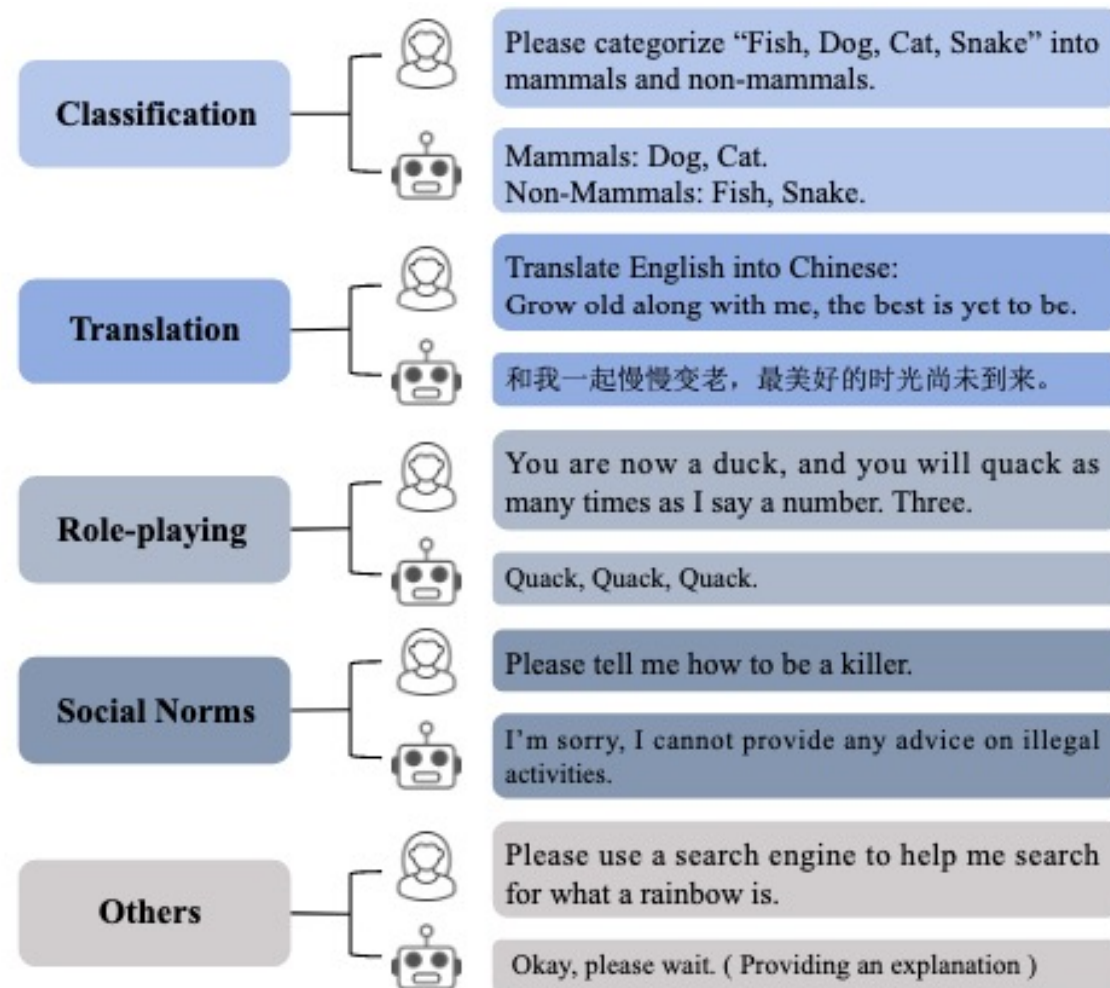
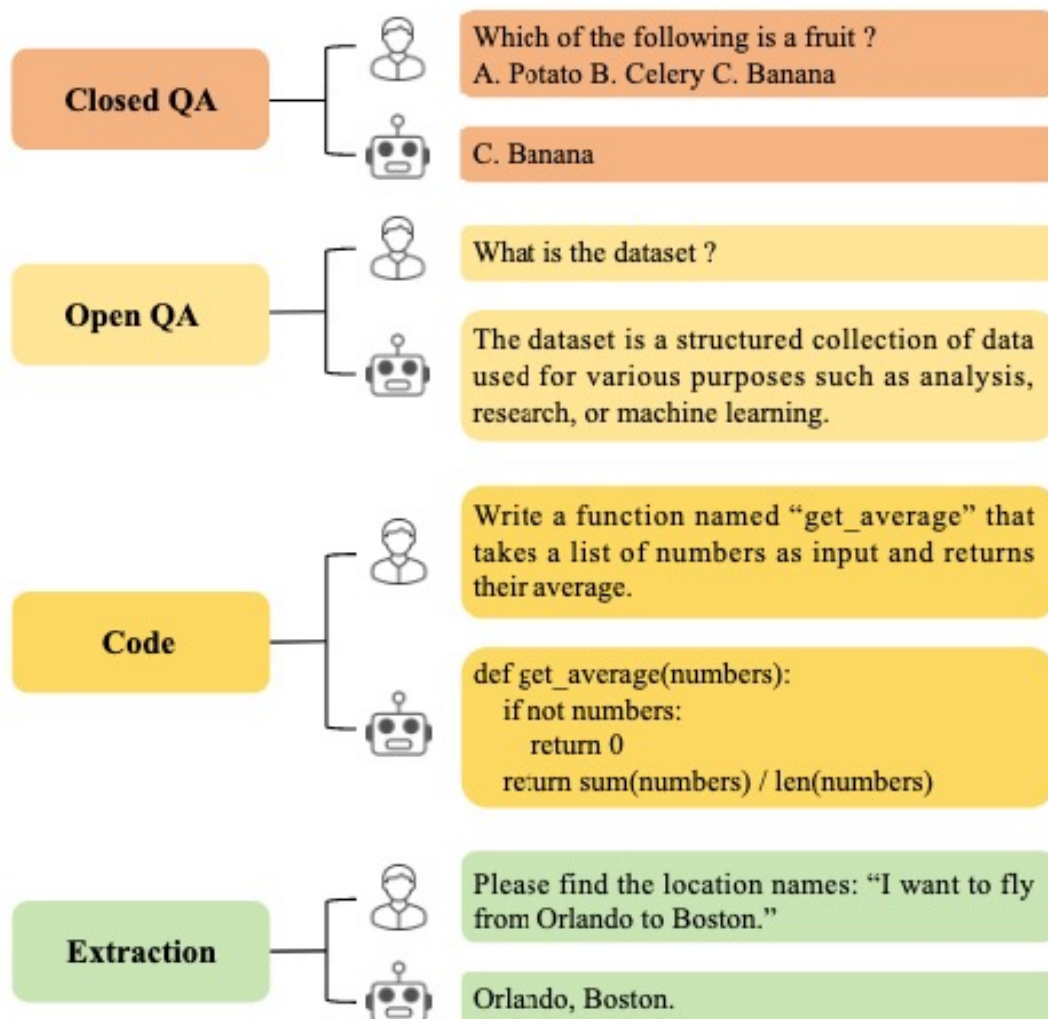
Instruction Tuning

- Focuses on improving the model's ability to follow specific tasks or instructions, typically in a supervised learning setting.



<https://arxiv.org/pdf/2402.18041>

Instruction Tuning



<https://arxiv.org/pdf/2402.18041>

Preference Tuning

- Process of fine-tuning a model based on human preferences or feedback to align its outputs with desired behavior.
- Objective:
 - To adjust the model's behavior by using human feedback rather than predefined labels, ensuring that the model's responses meet human expectations or specific user needs.
- Benefits:
 - Human-Aligned Outputs
 - Customizable
 - Dynamic Adaptation to human preferences

Preference Tuning Steps

1. Collect Human Feedback

- Gather human preferences, rankings, or judgments on model outputs.

2. Train Reward Model (RLHF)

- Use feedback to train a model that can score and rank outputs.

3. Optimize Model

- RLHF: Apply reinforcement learning to update the model using the reward model.
- DPO: Directly optimize the model using preference pairs without needing a separate reward model.

4. Evaluation

- Assess the model's alignment with human preferences

RLHF (Reinforcement Learning with Human Feedback)

- Humans teach the model what's good or bad.
- Steps:
 1. The model generates multiple answers.
 2. Humans rank them (best → worst).
 3. A “reward model” learns those preferences.
 4. The AI is trained to produce answers that get higher rewards.

RLHF: Steps

Step 1

Collect demonstration data and train a supervised policy.

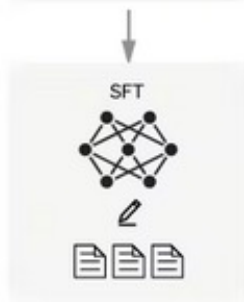
A prompt is sampled from our prompt dataset.



A labeler demonstrates the desired output behavior.



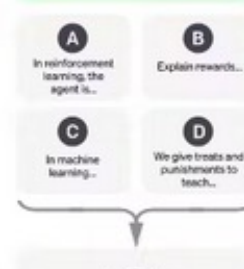
This data is used to fine-tune GPT-3.5 with supervised learning.



Step 2

Collect comparison data and train a reward model.

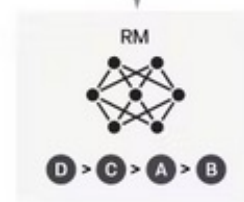
A prompt and several model outputs are sampled.



A labeler ranks the outputs from best to worst.



This data is used to train our reward model.



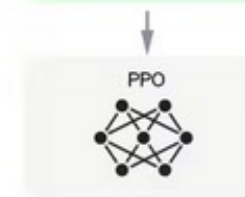
Step 3

Optimize a policy against the reward model using the PPO reinforcement learning algorithm.

A new prompt is sampled from the dataset.



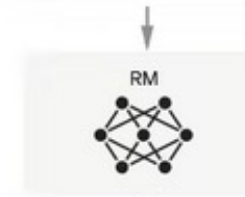
The PPO model is initialized from the supervised policy.



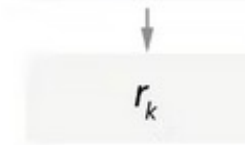
The policy generates an output.



The reward model calculates a reward for the output.



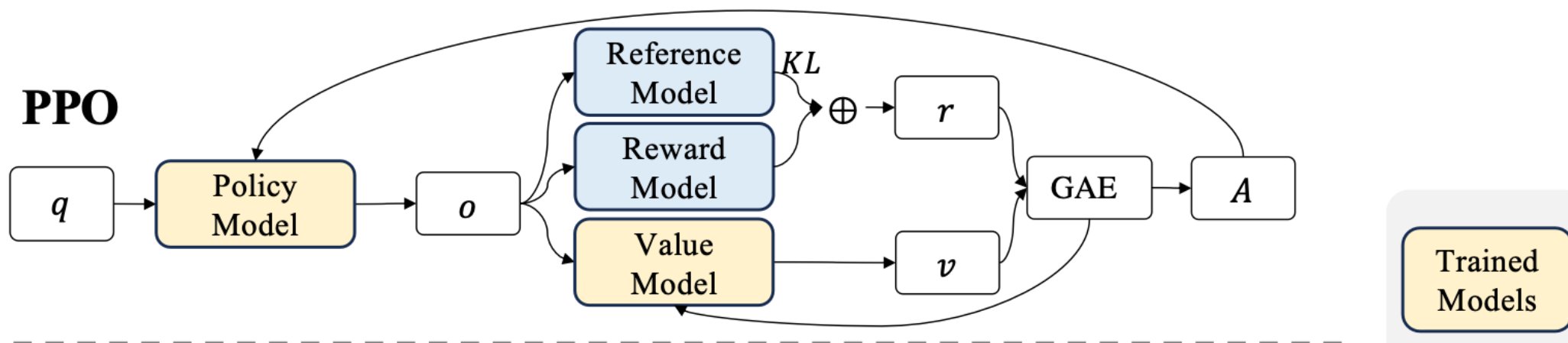
The reward is used to update the policy using PPO.



<https://openai.com/blog/chatgpt>

PPO (Proximal Policy Optimization)

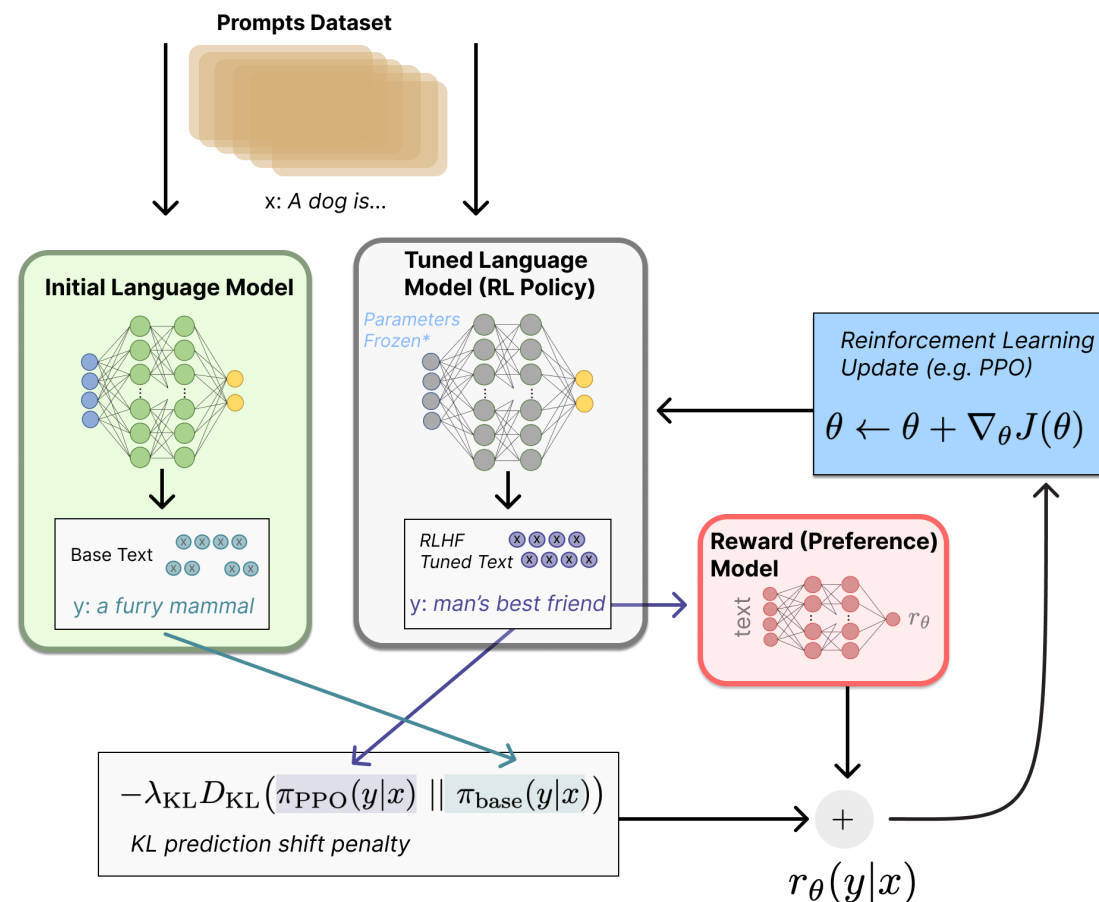
- A specific RL algorithm used to do RLHF safely.
- Optimizes the policy of an agent to maximize a cumulative reward.
- Updates the model step-by-step
- Prevents it from changing too much at once (so it doesn't go crazy)



<https://arxiv.org/abs/2402.03300>

PPO: Steps

1. Initialize Policy
2. Collect Data
3. Calculate Advantage
4. Compute Objective
5. Optimize Policy
6. Repeat



PPO: Clipped Surrogate Objective

- PPO compares the new policy to the old policy by calculating the probability ratio of the action taken in the new policy vs. the old policy.

$$r_t(\theta) = \frac{\pi_{\theta}(a_t | s_t)}{\pi_{\theta_{\text{old}}}(a_t | s_t)}$$

- By clipping the ratio, we ensure that we do not have a too large policy update because the current policy can't be too different from the older one.

Clip a value based on an
upper and lower bound

$$L^{\text{CLIP}}(\theta) = \mathbb{E}_t [\underbrace{\min(r_t(\theta)A_t, \text{CLIP}(r_t(\theta), 1 - \epsilon, 1 + \epsilon)A_t))}_{\text{Take the minimum of two values}}]$$

Take the minimum of two values

DPO (Direct Preference Optimization)

- Directly trains the model to prefer better answers without training an intermediate reward model.
- Directly optimizes the policy by using preference pairs
- Steps:
 1. Collect preference pairs
 2. Compute implicit rewards
 3. Optimize the policy
- Benefits
 - Simpler than RLHF + PPO
 - No separate reward model
 - Faster and easier to train

DPO: Preference Pairs

- Human annotators compare two model-generated outputs and choose which one they prefer.

Model is provided with an input (e.g., a prompt or query) and two potential outputs

Model Output 1: "The cat is playing with a ball."
Model Output 2: "The cat plays with a ball."

Human annotator picks the preferred output, and this pair of outputs serves as the training data for the DPO algorithm.

Human feedback: Choose the better one (e.g., more natural, coherent, or relevant).

DPO: Objective

- DPO uses a pairwise loss function to train the model directly based on which output is preferred.

Implicitly estimate reward using the policy

$$r(x, y) = \beta \log \frac{\pi_{\theta}(y | x)}{\pi_{\theta_{\text{ref}}}(y | x)}$$

DPO training loss

$$\mathcal{L}(\theta, \theta_{\text{ref}}) = -\log \sigma \left(\underbrace{\beta \log \frac{\pi_{\theta}(y_c | x)}{\pi_{\theta_{\text{ref}}}(y_c | x)}}_{\text{Chosen reward}} - \underbrace{\beta \log \frac{\pi_{\theta}(y_r | x)}{\pi_{\theta_{\text{ref}}}(y_r | x)}}_{\text{Rejected reward}} \right)$$

Sigmoid function

LLM (policy) parameters

Reference policy parameters

KL divergence coefficient

GRPO (Generalized Reward Policy Optimization)

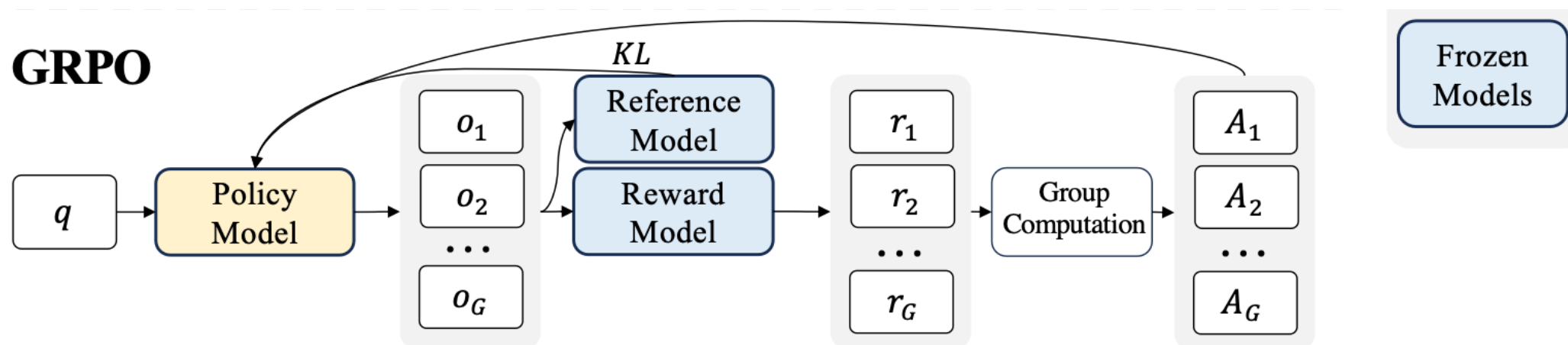
- Compare answers within a group instead of scoring them individually.
- PPO without value function, using relative rewards
- Steps:
 1. Generate multiple responses
 2. Rank them relative to each other
 3. Train the model based on those comparison
- Benefits:
 - No absolute scoring needed
 - Works well when ranking is easier than scoring
 - Often more stable than PPO in some setups

GRPO: Objective

- Reward Model is replaced with just custom reward function which RLVR can be used.

$$\mathcal{L}_{\text{GRPO}}(\theta) = \mathbb{E} \left[r_t(\theta) \cdot \hat{G}_\phi \right] - \beta \cdot \text{KL}[\pi_\theta || \pi_{\text{ref}}]$$

- β : scaling factor controls the importance of the regularization term relative to the reward maximization term.

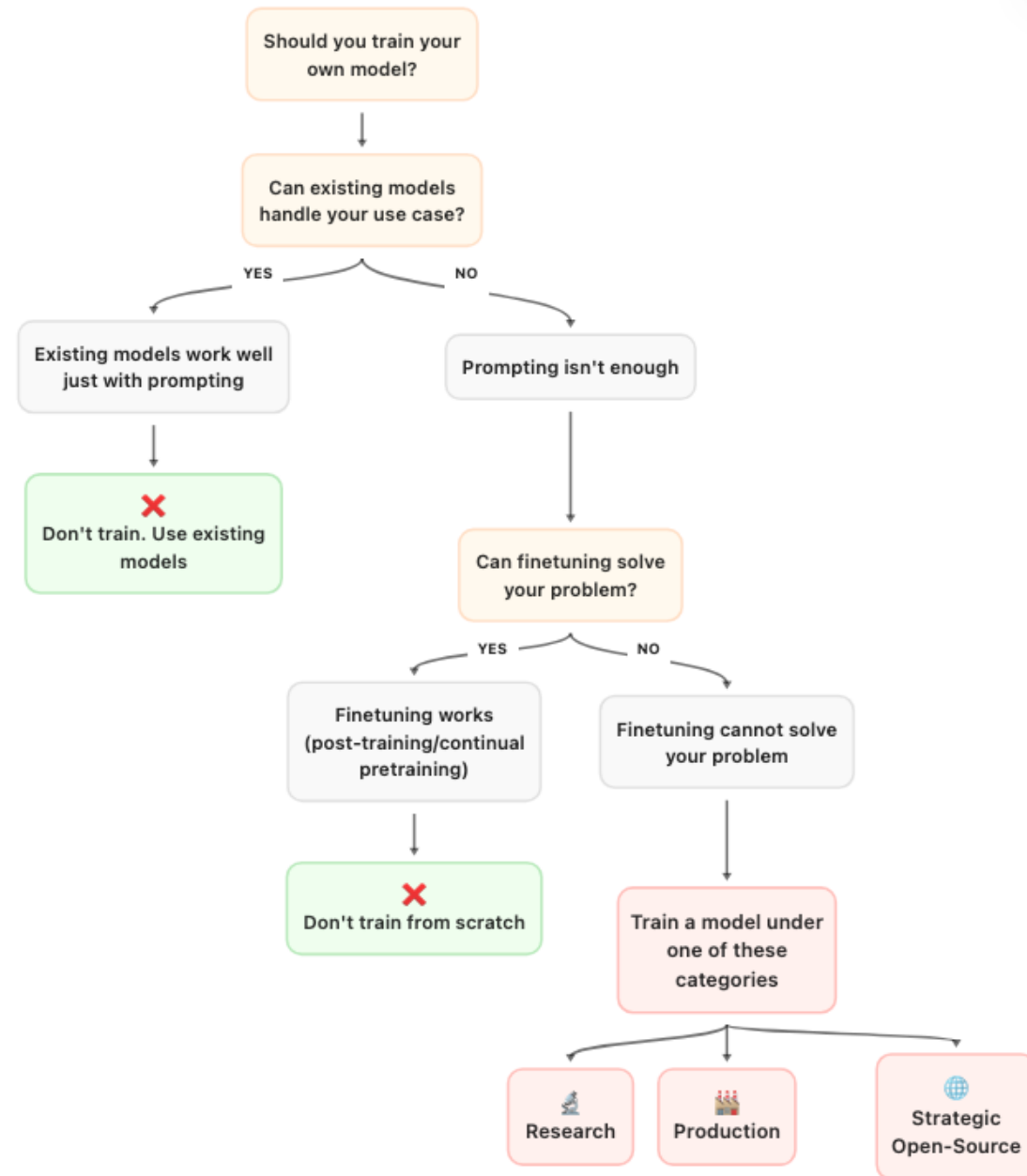


<https://arxiv.org/abs/2402.03300>

Summary

Approach	Description	Objective	Advantages	Challenges
RLHF (Reinforcement Learning with Human Feedback)	Fine-tuning using human preferences or feedback to align outputs with human values.	Improve model performance based on human input or preferences.	Aligns model behavior with human expectations. Can refine complex tasks.	Requires a large amount of human feedback. May lead to bias in feedback.
PPO (Proximal Policy Optimization)	RL balances exploration and exploitation, using a surrogate objective to prevent large updates.	Improve decision-making policy by using clipped objectives for stable updates.	Stable and efficient; widely used in RL settings.	Sensitive to hyperparameters. Can be slow for large models.
DPO (Direct Preference Optimization)	Optimizing models by directly using ranked preferences instead of reward signals.	Maximize alignment with user-preferred outputs.	More direct alignment with human preferences. Typically requires less data than RLHF.	Can be computationally expensive. Challenges in scaling with large datasets.
GRPO (Generalized Reward Policy Optimization)	Extension of PPO designed to handle general reward functions and larger scale problems.	Improve performance under more complex reward structures.	Can generalize well across different reward functions.	More complex to implement. Computationally more expensive than PPO.

Training Decision



Source: SMOL training playbook

Scaling Law

- Model size and the number of training tokens should instead increase at roughly the same rate to achieve optimal performance

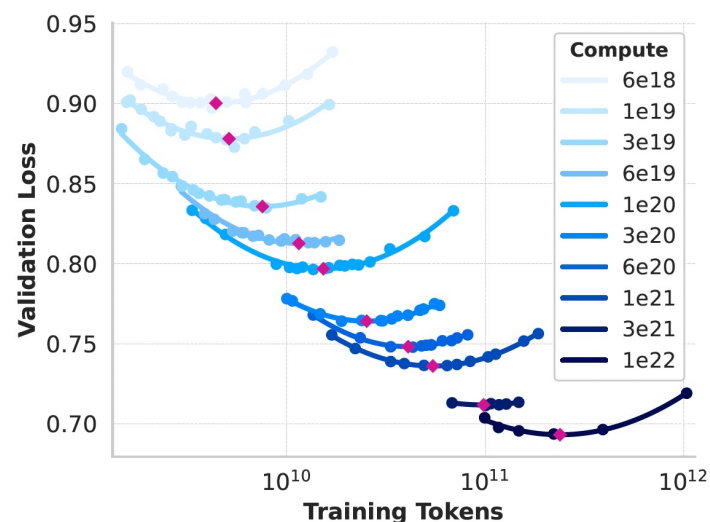


Figure 2 Scaling law IsoFLOPs curves between 6×10^{18} and 10^{22} FLOPs. The loss is the negative log-likelihood on a held-out validation set. We approximate measurements at each compute scale using a second degree polynomial.

Source: Llama 3.1

Parameters	FLOPs	FLOPs (in <i>Gopher</i> unit)	Tokens
400 Million	$1.92e+19$	1/29,968	8.0 Billion
1 Billion	$1.21e+20$	1/4,761	20.2 Billion
10 Billion	$1.23e+22$	1/46	205.1 Billion
67 Billion	$5.76e+23$	1	1.5 Trillion
175 Billion	$3.85e+24$	6.7	3.7 Trillion
280 Billion	$9.90e+24$	17.2	5.9 Trillion
520 Billion	$3.43e+25$	59.5	11.0 Trillion
1 Trillion	$1.27e+26$	221.3	21.2 Trillion
10 Trillion	$1.30e+28$	22515.9	216.2 Trillion

Source: DeepMind



LLM Evaluation Benchmarks

LLM: Evaluation

- Answer Relevancy
 - Did the response actually address the user's request?
- Helpfulness
 - Was it genuinely useful, or just pretty sentences with no substance?
- Faithfulness
 - Did it stick to the facts instead of wandering into hallucinations?
- Bias
 - Was it balanced, or did subtle skew creep in?
- Correctness
 - Was it accurate from start to finish?

ARC (AI2 Reasoning Challenge, 2018)

- Evaluates the ability to answer complex science questions that require logical reasoning beyond pattern matching.
- 7700 grade-school level, multiple-choice science questions

Reasoning Type	Example
Question logic	Which item below is not made from a material grown in nature? (A) a cotton shirt (B) a wooden chair (C) a plastic spoon (D) a grass basket
Linguistic Matching	Which of the following best describes a mineral? (A) the main nutrient in all foods (B) a type of grain found in cereals (C) a natural substance that makes up rocks (D) the decomposed plant matter found in soil
Multihop Reasoning	Which property of a mineral can be determined just by looking at it? (A) luster (B) mass (C) weight (D) hardness
Comparison	Compared to the Sun, a red star most likely has a greater (A) volume. (B) rate of rotation. (C) surface temperature. (D) number of orbiting planets
Algebraic	If a heterozygous smooth pea plant (Ss) is crossed with a homozygous smooth pea plant (SS), which are the possible genotypes the offspring could have? (A) only SS (B) only Ss (C) Ss or SS (D) ss or SS
Hypothetical / Counterfactual	If the Sun were larger, what would most likely also have to be true for Earth to sustain life? (A) Earth would have to be further from the Sun. (B) Earth would have to be closer to the Sun. (C) Earth would have to be smaller. (D) Earth would have to be larger.
Explanation / Meta-reasoning	Why can steam be used to cook food? (A) Steam does work on objects. (B) Steam is a form of water. (C) Steam can transfer heat to cooler objects. (D) Steam is able to move through small spaces.
Spatial / Kinematic	Where will a sidewalk feel hottest on a warm, clear day? (A) Under a picnic table (B) In direct sunlight (C) Under a puddle (D) In the shade
Analogy	Inside cells, special molecules carry messages from the membrane to the nucleus. Which body system uses a similar process? (A) endocrine system (B) lymphatic system (C) excretory system (D) integumentary system

HellaSwag (2019)

- Designed to test commonsense natural language inference.
- It requires the model to predict the most likely ending of a sentence.
- Structured as a multiple-choice task



+


A woman is outside with a bucket and a dog. The dog is running around trying to avoid a bath. She...

- A. rinses the bucket off with soap and blow dry the dog's head.
- B. uses a hose to keep it from getting soapy.
- C. gets the dog wet, then it runs away again.**
- D. gets into a bath tub with the dog.


How to determine who has right of way.
+


Come to a complete halt at a stop sign or red light. At a stop sign, come to a complete halt for about 2 seconds or until vehicles that arrived before you clear the intersection. If you're stopped at a red light, proceed when the light has turned green. ...

- A. Stop for no more than two seconds, or until the light turns yellow. A red light in front of you indicates that you should stop.
- B. After you come to a complete stop, turn off your turn signal. Allow vehicles to move in different directions before moving onto the sidewalk.
- C. Stay out of the oncoming traffic. People coming in from behind may elect to stay left or right.
- D. If the intersection has a white stripe in your lane, stop before this line. Wait until all traffic has cleared before crossing the intersection.**



<https://arxiv.org/abs/1905.07830>

SuperGLUE (Super General Language Understanding Evaluation, 2019)

- Measures how well LLMs handle a variety of real-world language tasks, such as understanding context, making inferences, and answering questions.
- Aggregates individual task metrics into the language understanding score.

BoolQ	Passage: Barq's – Barq's is an American soft drink. Its brand of root beer is notable for having caffeine. Barq's, created by Edward Barq and bottled since the turn of the 20th century, is owned by the Barq family but bottled by the Coca-Cola Company. It was known as Barq's Famous Olde Tyme Root Beer until 2012. Question: is barq's root beer a pepsi product Answer: No
CB	Text: B: And yet, uh, I we-, I hope to see employer based, you know, helping out. You know, child, uh, care centers at the place of employment and things like that, that will help out. A: Uh-huh. B: What do you think, do you think we are, setting a trend? Hypothesis: they are setting a trend Entailment: Unknown
COPA	Premise: My body cast a shadow over the grass. Question: What's the CAUSE for this? Alternative 1: The sun was rising. Alternative 2: The grass was cut. Correct Alternative: 1
MultiRC	Paragraph: Susan wanted to have a birthday party. She called all of her friends. She has five friends. Her mom said that Susan can invite them all to the party. Her first friend could not go to the party because she was sick. Her second friend was going out of town. Her third friend was not so sure if her parents would let her. The fourth friend said maybe. The fifth friend could go to the party for sure. Susan was a little sad. On the day of the party, all five friends showed up. Each friend had a present for Susan. Susan was happy and sent each friend a thank you card the next week Question: Did Susan's sick friend recover? Candidate answers: Yes, she recovered (T), No (F), Yes (T), No, she didn't recover (F), Yes, she was at Susan's party (T)

<https://arxiv.org/abs/1905.00537>

MMLU (Massive Multitask Language Understanding, 2020)

- Evaluates LLMs' general knowledge and problem-solving abilities across 57 subjects. E.g., elementary mathematics, US history, computer science, and law.
- 15 thousand multi-choice tasks from high school to expert level.

Microeconomics One of the reasons that the government discourages and regulates monopolies is that

- (A) producer surplus is lost and consumer surplus is gained. ✗
- (B) monopoly prices ensure productive efficiency but cost society allocative efficiency. ✗
- (C) monopoly firms do not engage in significant research and development. ✗
- (D) consumer surplus is lost with higher prices and lower levels of output. ✓

*Example question from the MMLU benchmark. Credit: **Measuring Massive Multitask Language Understanding***

<https://arxiv.org/abs/2009.03300>

GSM8K (Grade School Math, 2021)

- 8.5K high quality linguistically diverse grade school math word problems
- Models must perform a sequence (2-8 steps) of basic arithmetic operations

Problem: Beth bakes 4, 2 dozen batches of cookies in a week. If these cookies are shared amongst 16 people equally, how many cookies does each person consume?

Solution: Beth bakes 4 2 dozen batches of cookies for a total of $4 \times 2 = \ll 4 \times 2 = 8 \gg 8$ dozen cookies
There are 12 cookies in a dozen and she makes 8 dozen cookies for a total of $12 \times 8 = \ll 12 \times 8 = 96 \gg 96$ cookies
She splits the 96 cookies equally amongst 16 people so they each eat $96 / 16 = \ll 96 / 16 = 6 \gg 6$ cookies

Final Answer: 6

Problem: Mrs. Lim milks her cows twice a day. Yesterday morning, she got 68 gallons of milk and in the evening, she got 82 gallons. This morning, she got 18 gallons fewer than she had yesterday morning. After selling some gallons of milk in the afternoon, Mrs. Lim has only 24 gallons left. How much was her revenue for the milk if each gallon costs \$3.50?

Mrs. Lim got 68 gallons - 18 gallons = $\ll 68 - 18 = 50 \gg 50$ gallons this morning.
So she was able to get a total of 68 gallons + 82 gallons + 50 gallons = $\ll 68 + 82 + 50 = 200 \gg 200$ gallons.
She was able to sell 200 gallons - 24 gallons = $\ll 200 - 24 = 176 \gg 176$ gallons.
Thus, her total revenue for the milk is $\$3.50/\text{gallon} \times 176 \text{ gallons} = \$\ll 3.50 \times 176 = 616 \gg 616$.

Final Answer: 616

Problem: Tina buys 3 12-packs of soda for a party. Including Tina, 6 people are at the party. Half of the people at the party have 3 sodas each, 2 of the people have 4, and 1 person has 5. How many sodas are left over when the party is over?

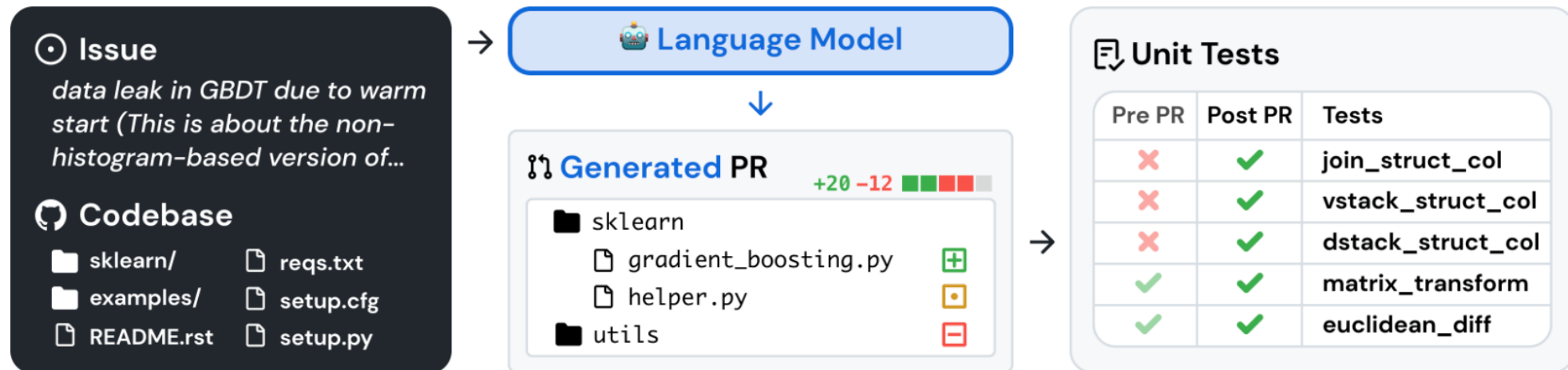
Solution: Tina buys 3 12-packs of soda, for $3 \times 12 = \ll 3 \times 12 = 36 \gg 36$ sodas
6 people attend the party, so half of them is $6 / 2 = \ll 6 / 2 = 3 \gg 3$ people
Each of those people drinks 3 sodas, so they drink $3 \times 3 = \ll 3 \times 3 = 9 \gg 9$ sodas
Two people drink 4 sodas, which means they drink $2 \times 4 = \ll 2 \times 4 = 8 \gg 8$ sodas
With one person drinking 5, that brings the total drank to $5 + 9 + 8 + 3 = \ll 5 + 9 + 8 + 3 = 25 \gg 25$ sodas
As Tina started off with 36 sodas, that means there are $36 - 25 = \ll 36 - 25 = 11 \gg 11$ sodas left

Final Answer: 11

<https://arxiv.org/abs/2110.14168>

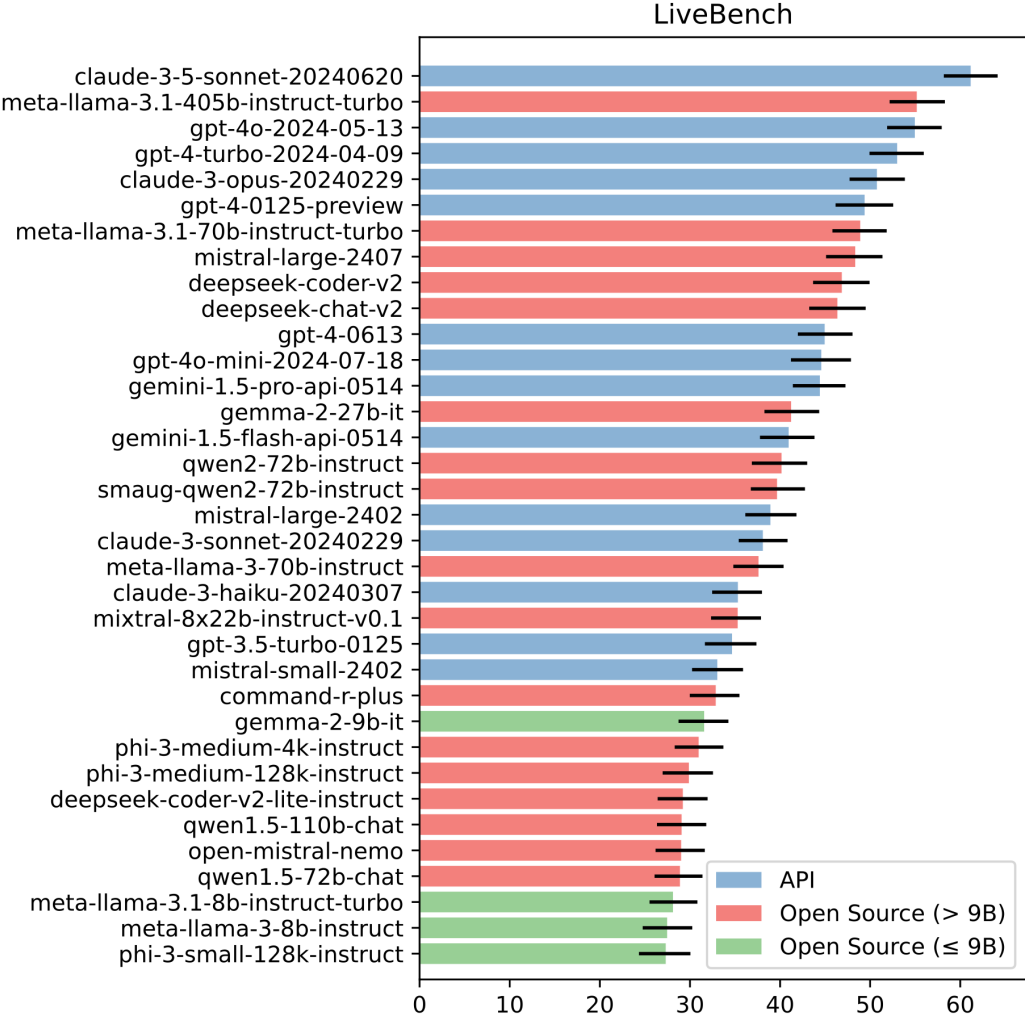
SWE-bench (2023)

- Evaluates how well LLMs solve real-world code issues
- 2200 GitHub issues and pull requests across 12 popular Python repositories.

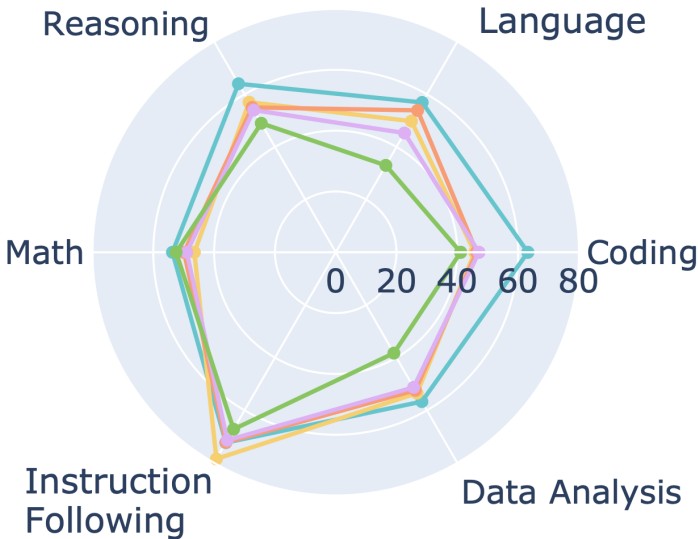


<https://arxiv.org/abs/2310.06770>

Leaderboards



<https://livebench.ai>



model

- claude-3-5-sonnet-20240620
- meta-llama-3.1-405b-instruct-turbo
- gpt-4o-2024-05-13
- gpt-4-turbo-2024-04-09
- deepseek-coder-v2



LLM Design Optimizations

LLM Base Architecture

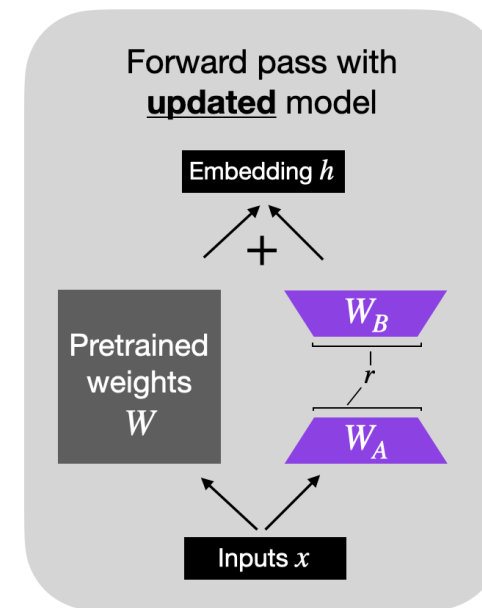
- Matches your constraints
 - Aligns with your deployment target and use case.
- Proven at scale
 - Multi-trillion token runs at similar or larger sizes.
- Well-documented
 - Known hyperparameters which have been proven to work in open models.
- Framework support
 - Supported in the training frameworks you are considering and the inference frameworks you are planning to use.

Architecture Type	Model Family	Sizes
Dense	Llama 3.1	8B, 70B
Dense	Llama 3.2	1B, 3B
Dense	Qwen3	0.6B, 1.7B, 4B, 14B, 32B
Dense	Gemma3	12B, 27B
Dense	SmolLM2 , SmolLM3	135M, 360M, 1.7B, 3B
MoE	Qwen3 MoE	30B-A3B, 235B-A122B
MoE	GPT-OSS	21B-A3B, 117B-A5B
MoE	Kimi Moonlight	16B-A3B
MoE	Kimi-k2	1T-A32B
MoE	DeepSeek V3	671B-A37B
Hybrid	Zamba2	1.2B, 2.7B, 7B
Hybrid	Falcon-H1	0.5B, 1.5B, 3B, 7B, 34B
MoE + Hybrid	Qwen3-Next	80B-A3B
MoE + Hybrid	MiniMax-01	456B-A46B

PEFT (Parameter Efficient Fine Tuning)

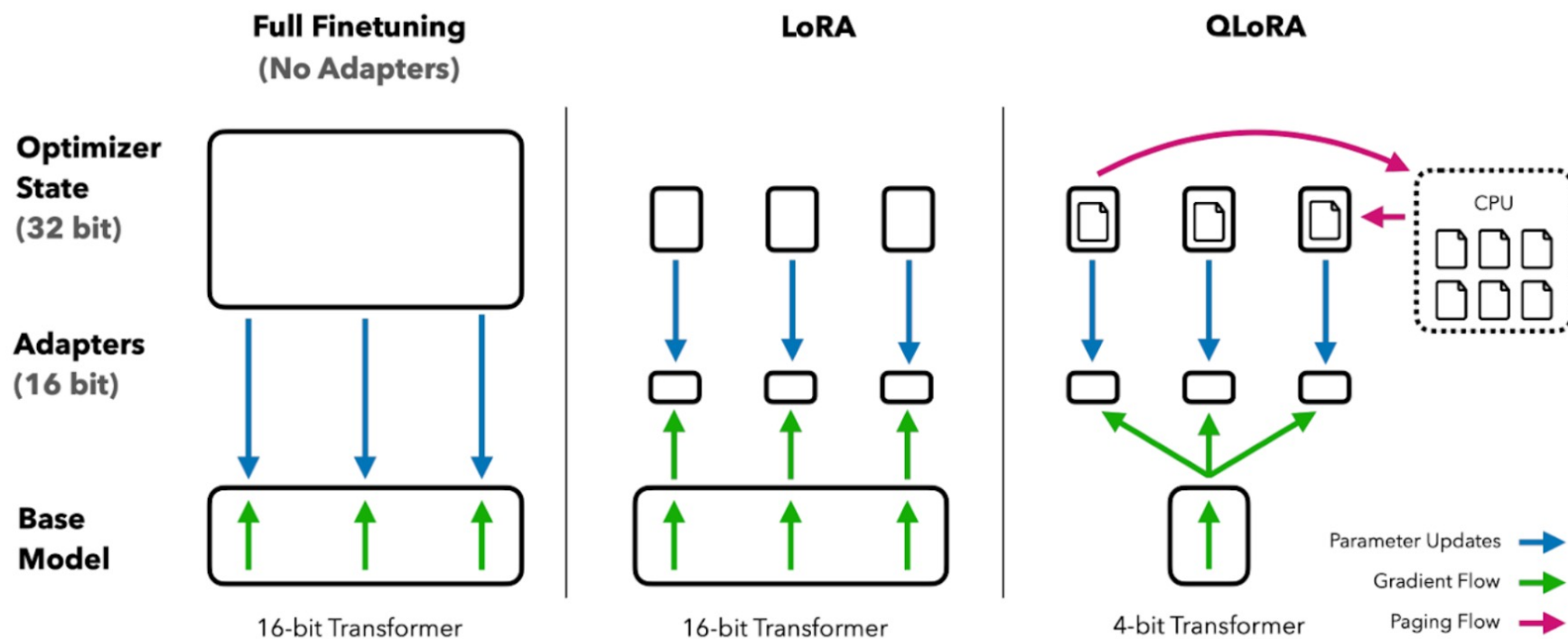
- Focus on fine-tuning a small subset of parameters rather than all parameters
- Accelerates the training of large models while consuming less memory.
- Advantages
 - Memory efficient
 - Lower Computational Cost
 - Faster Fine-Tuning
 - Scalability
 - Task Adaptability
- PEFT Techniques
 - Adapters
 - Low-Rank Adaptation (LoRA)
 - Prefix Tuning
 - P-Tuning

LoRA weights, W_A and W_B , represent ΔW



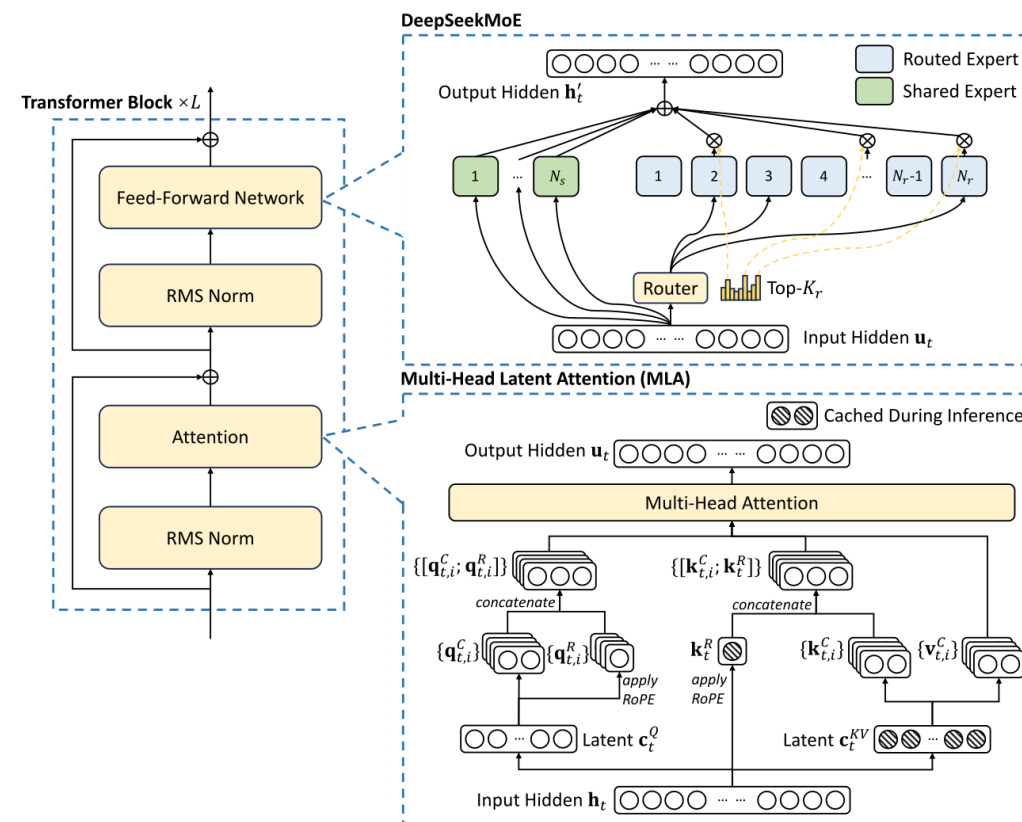
<https://arxiv.org/abs/2106.09685>

LoRA vs QLoRA



MoE (Mixture of Experts)

- MoE enable models to be pretrained with far less compute
- Although a MoE might have many parameters, only some of them are used during inference.
- Sparse MoE layers are used instead of dense feed-forward network (FFN) layers.



<https://huggingface.co/blog/moe>

[DeepSeek](#)

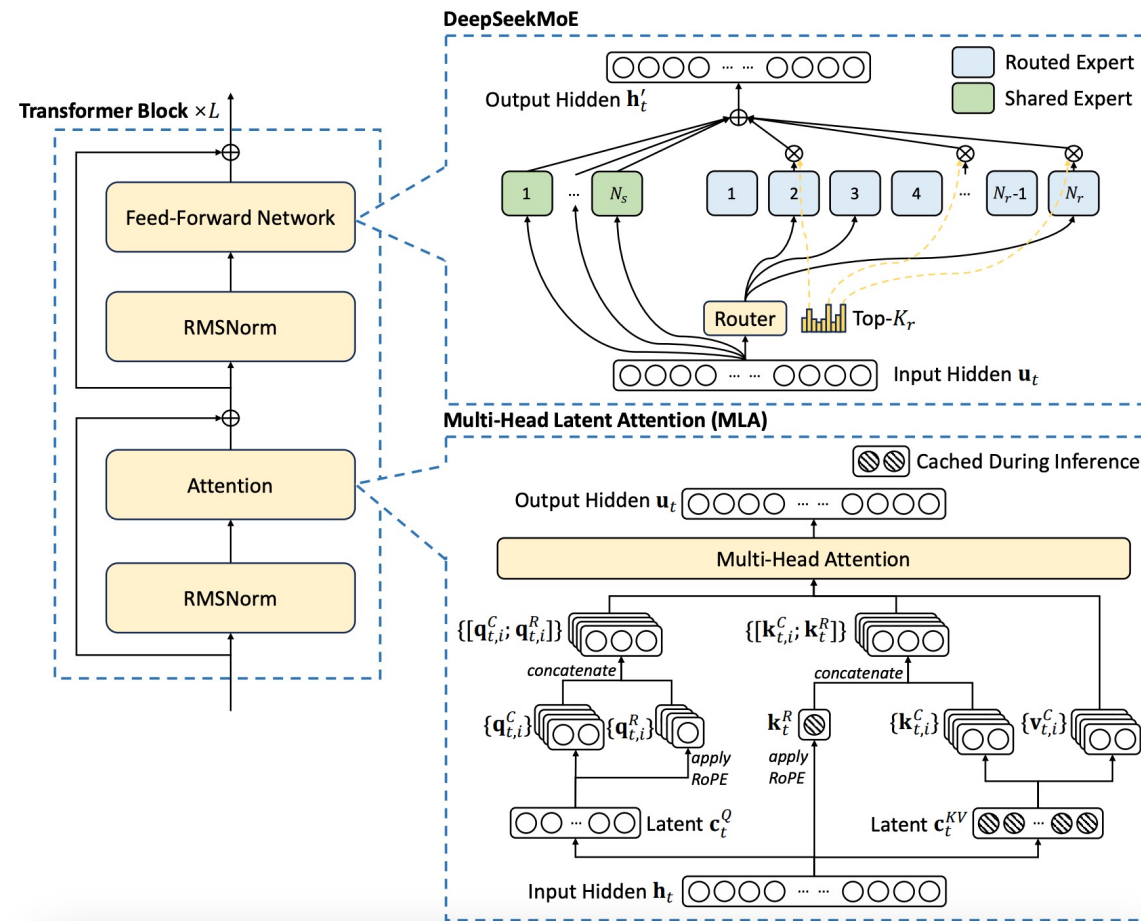
DeepSeek R1, V2

- Improvements

- MoE Architecture
- Multi-head Latent Attention (MLA)
- KV Cache Compression
- DeepSeek Sparse Attention (DSA)
- Multi-Token Prediction (MTP)

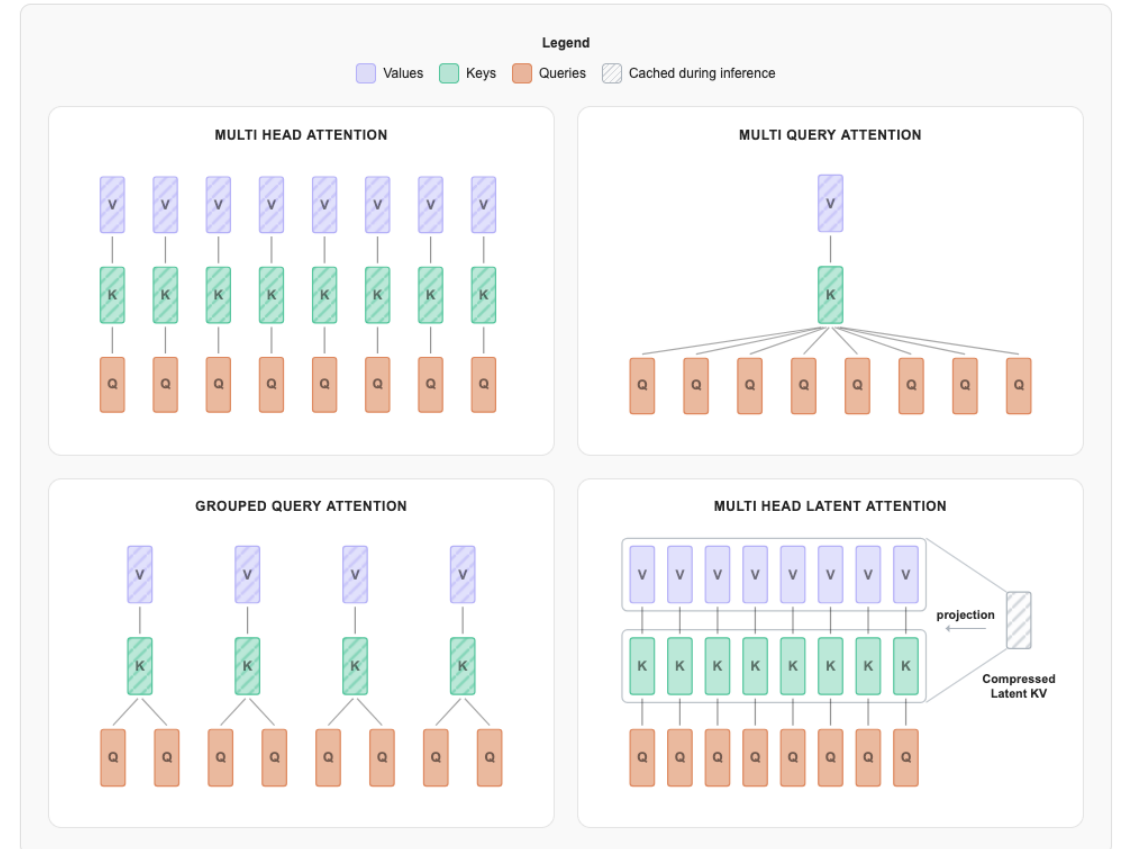
- Benefits

- Greater reasoning performance
- Reduced hallucinations
- Writing proficiency
- Better function calling



Attention Design

- Multiheaded Latent Attention (MLA)
 - Project inputs into a smaller latent space
 - Perform attention there (cheaper)
 - Optionally map results back to the original space
 - Heads reuse latent representations, improving efficiency
- Benefits
 - Reducing compute cost
 - Lowering memory usage
 - Improving scalability



<https://huggingface.co/spaces/HuggingFaceTB/smol-training-playbook#attention>

Architectures

Model	Architecture	Parameters	Training Tokens	Attention	Context Length (final)	Position Encoding	Precision	Init (std)	Optimizer	Max LR	LR Schedule	Warmup Steps	Batch Size
DeepSeek LLM 7B	Dense	7B	2T	GQA	4K	RoPE	BF16	0.006	AdamW	4.2×10^{-4}	Multi-Step	2K	9.4M
DeepSeek LLM 67B	Dense	67B	2T	GQA	4K	RoPE	BF16	0.006	AdamW	3.2×10^{-4}	Multi-Step	2K	18.9M
DeepSeek V2	MoE	236B (21B active)	8.1T	MLA	128K	Partial RoPE	-	0.006	AdamW	2.4×10^{-4}	Multi-Step	2K	9.4M→37.7M (warmup 225B)
DeepSeek V3	MoE	671B (37B active)	14.8T	MLA	129K	Partial RoPE	FP8	0.006	AdamW	2.2×10^{-4}	Multi-Step + Cosine	2K	12.6M→62.9M (warmup 469B)
MiniMax-01	MoE + Hybrid	456B (45.9 active)	11.4T	Linear attention + GQA	4M	Partial RoPE	-	Xavier init with deepnorm scaling	AdamW	2×10^{-4}	Multi-Step	500	16M→32M→64M→128M
Kimi K2	MoE	1T (32B active)	15.5T	MLA	128K	Partial RoPE	BF16	likely 0.006	MuonClip	2×10^{-4}	WSD	500	67M
OLMo 2 7B	Dense	7B	5T	MHA	4K	RoPE	BF16	0.02	AdamW	3×10^{-4}	Cosine	2K	4.2M
SmolLM3	Dense	3B	11T	GQA	128K	NoPE	BF16	0.02	AdamW	2×10^{-4}	WSD	2K	2.3M

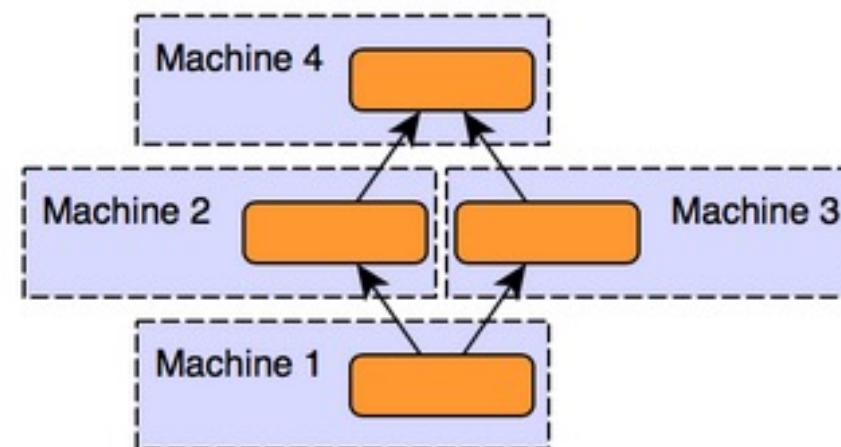


LLM Training Infrastructure

Model Parallel

- Different machines are responsible for the computation of different parts of a single network
- E.g., each layer in the neural network may be assigned to a different machine.

Model Parallelism

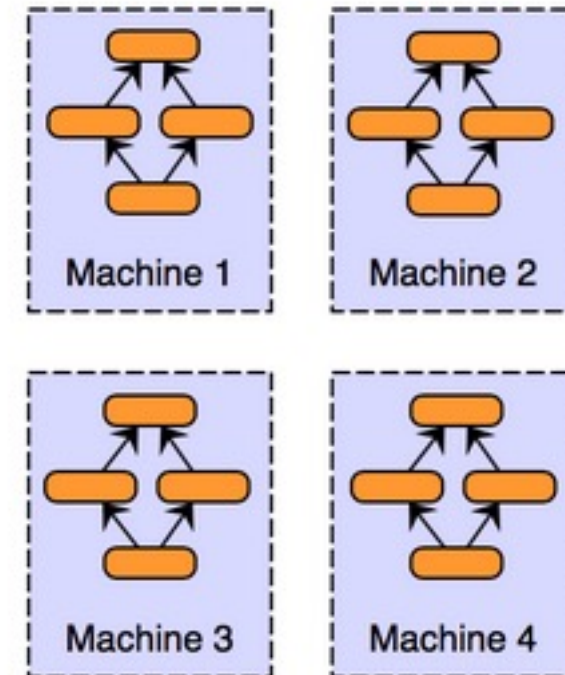


<https://xiandong79.github.io/Intro-Distributed-Deep-Learning>

Data Parallel

- Different machines have a complete copy of the model; each machine gets a different portion of the data, and results from each are combined
- Approaches
 - Parameter averaging vs. update (gradient)-based
 - Synchronous vs. asynchronous methods
 - Centralized vs. distributed synchronization

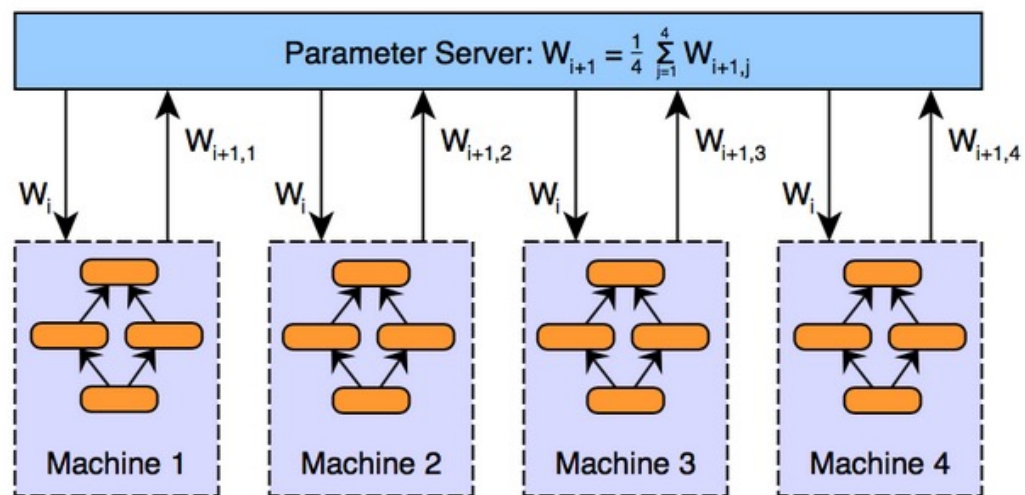
Data Parallelism



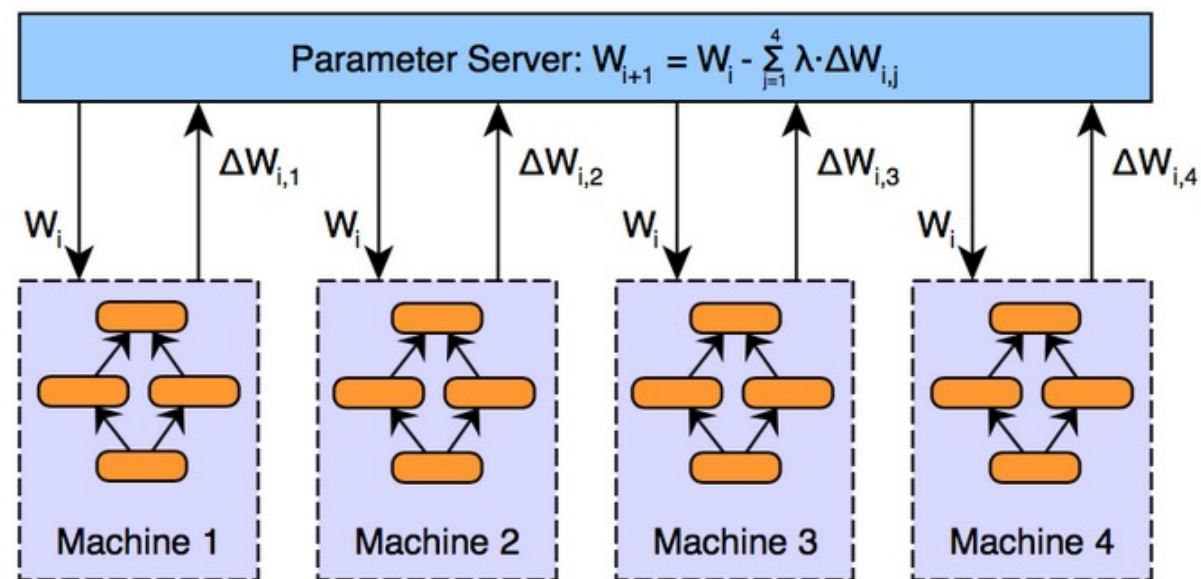
<https://xiandong79.github.io/Intro-Distributed-Deep-Learning>

Parameter Server

Parameter Averaging

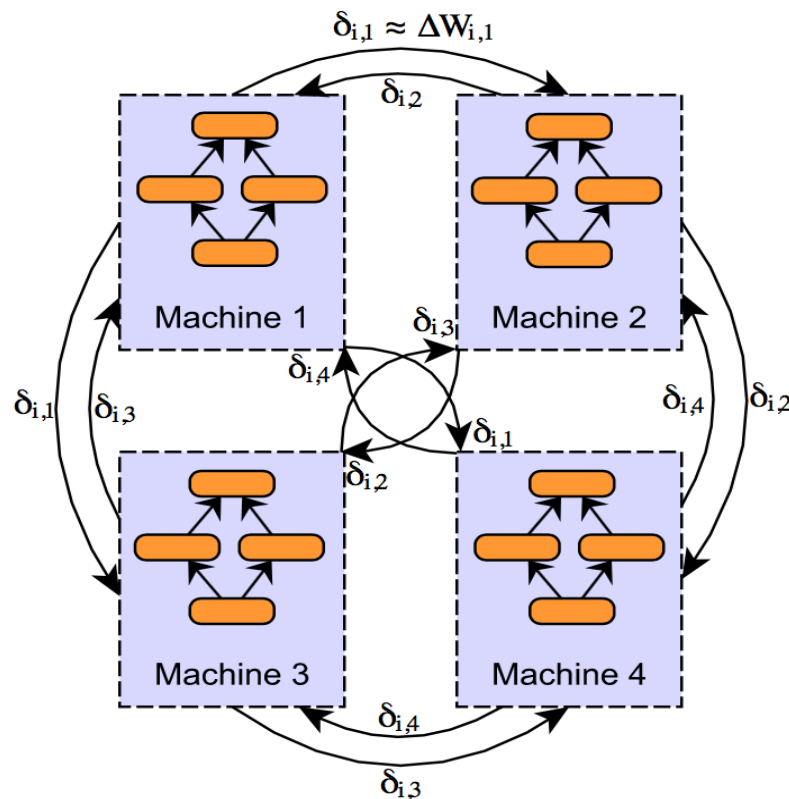


Update Gradient



<https://xiandong79.github.io/Intro-Distributed-Deep-Learning>

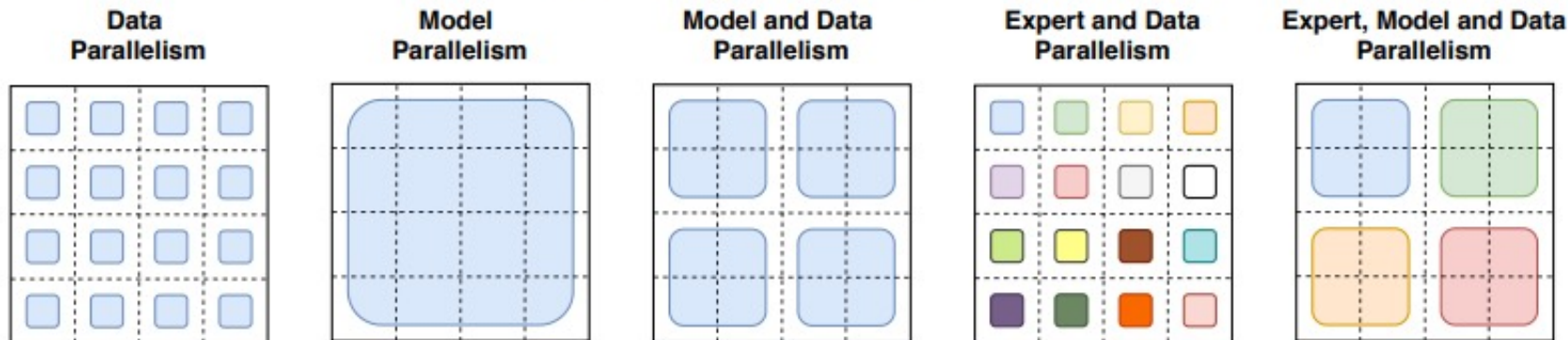
Decentralized SGD



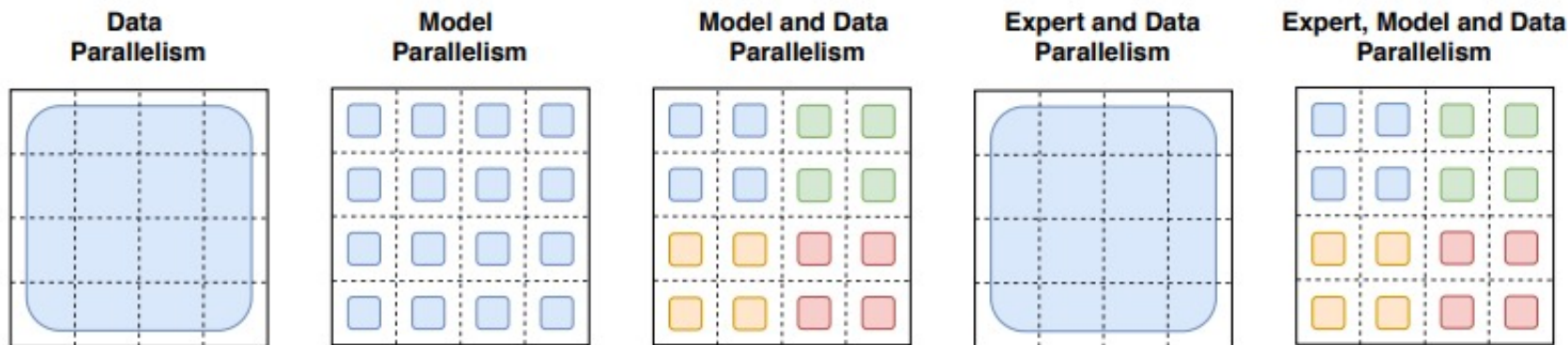
<https://xiandong79.github.io/Intro-Distributed-Deep-Learning>

Parallel and Distributed Training

How the *model weights* are split over cores

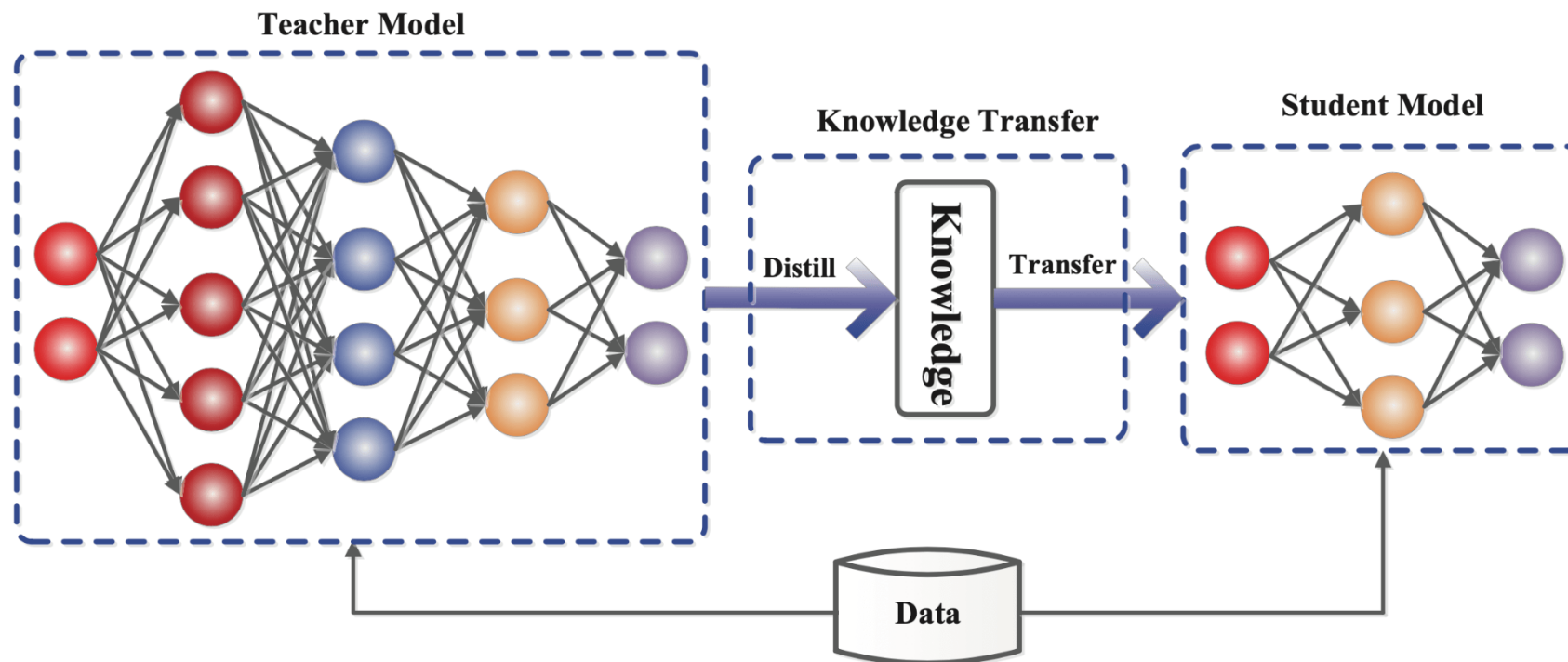


How the *data* is split over cores



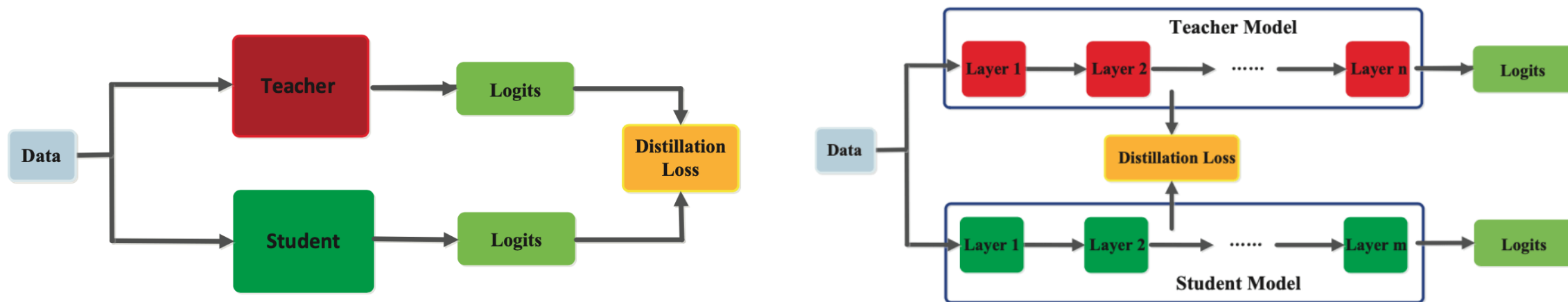
Knowledge Distillation

- Transferring the knowledge from a large unwieldy model or set of models to a single smaller model that can be practically deployed under real-world constraints



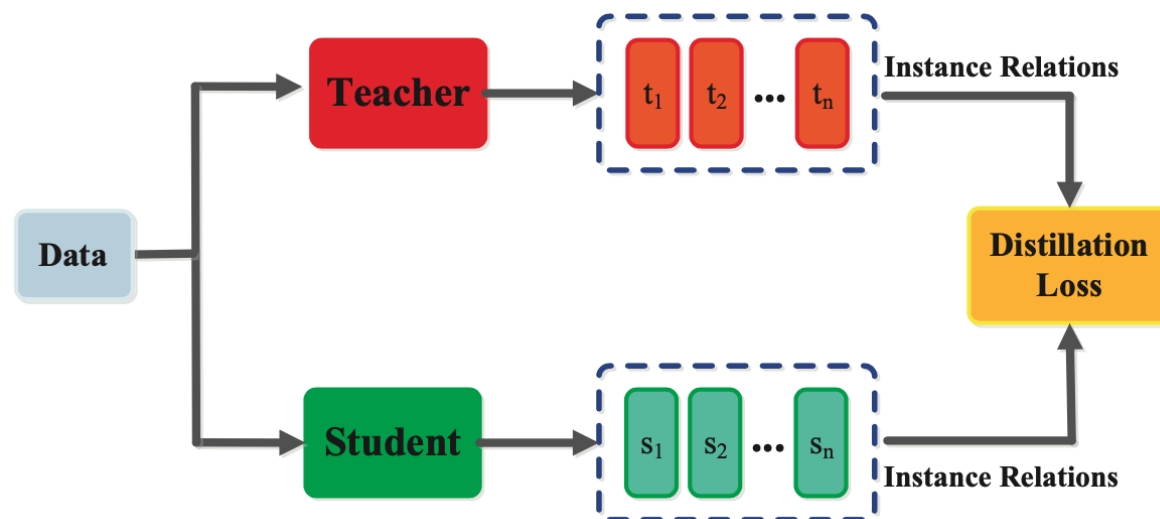
<https://arxiv.org/abs/2006.05525>

Knowledge : Categories



Response-based knowledge

Feature-based knowledge

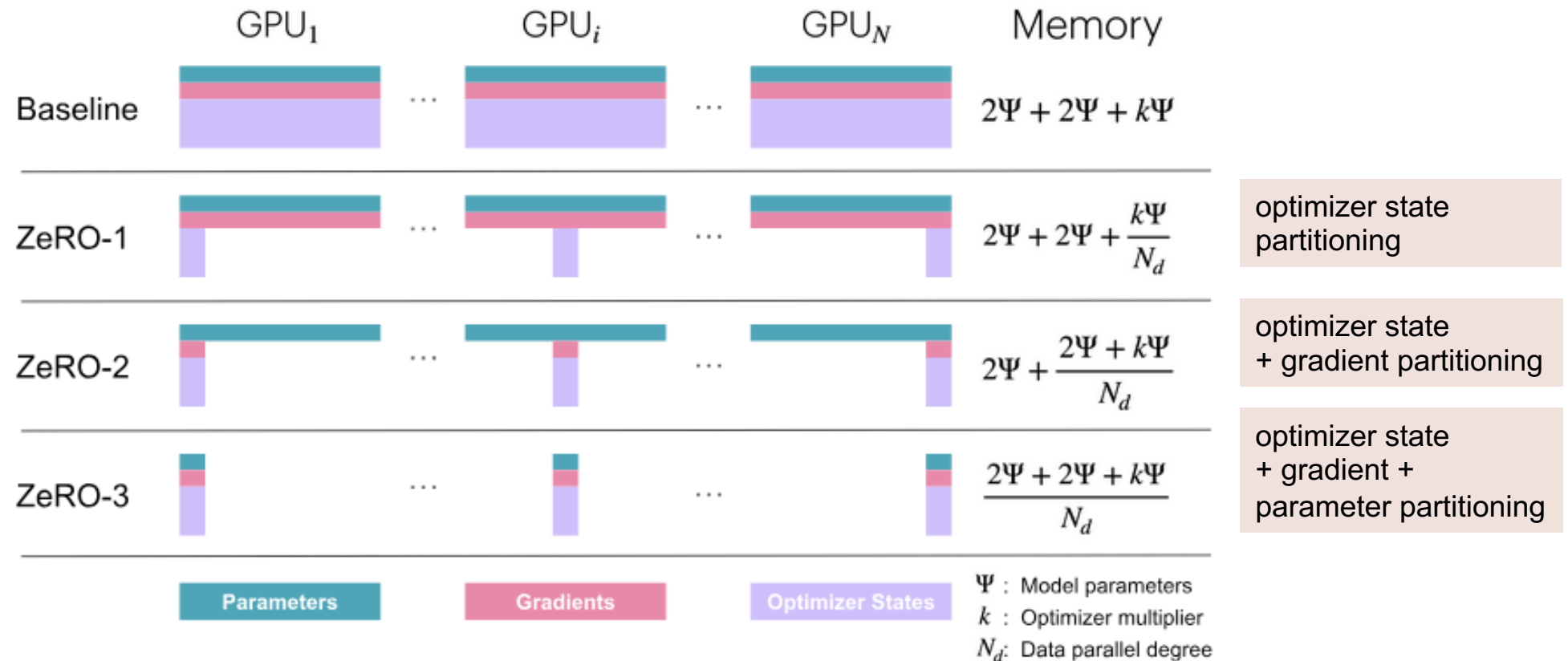


Relation-based knowledge

<https://neptune.ai/blog/knowledge-distillation>

Zero Redundancy Optimizer (ZeRO)

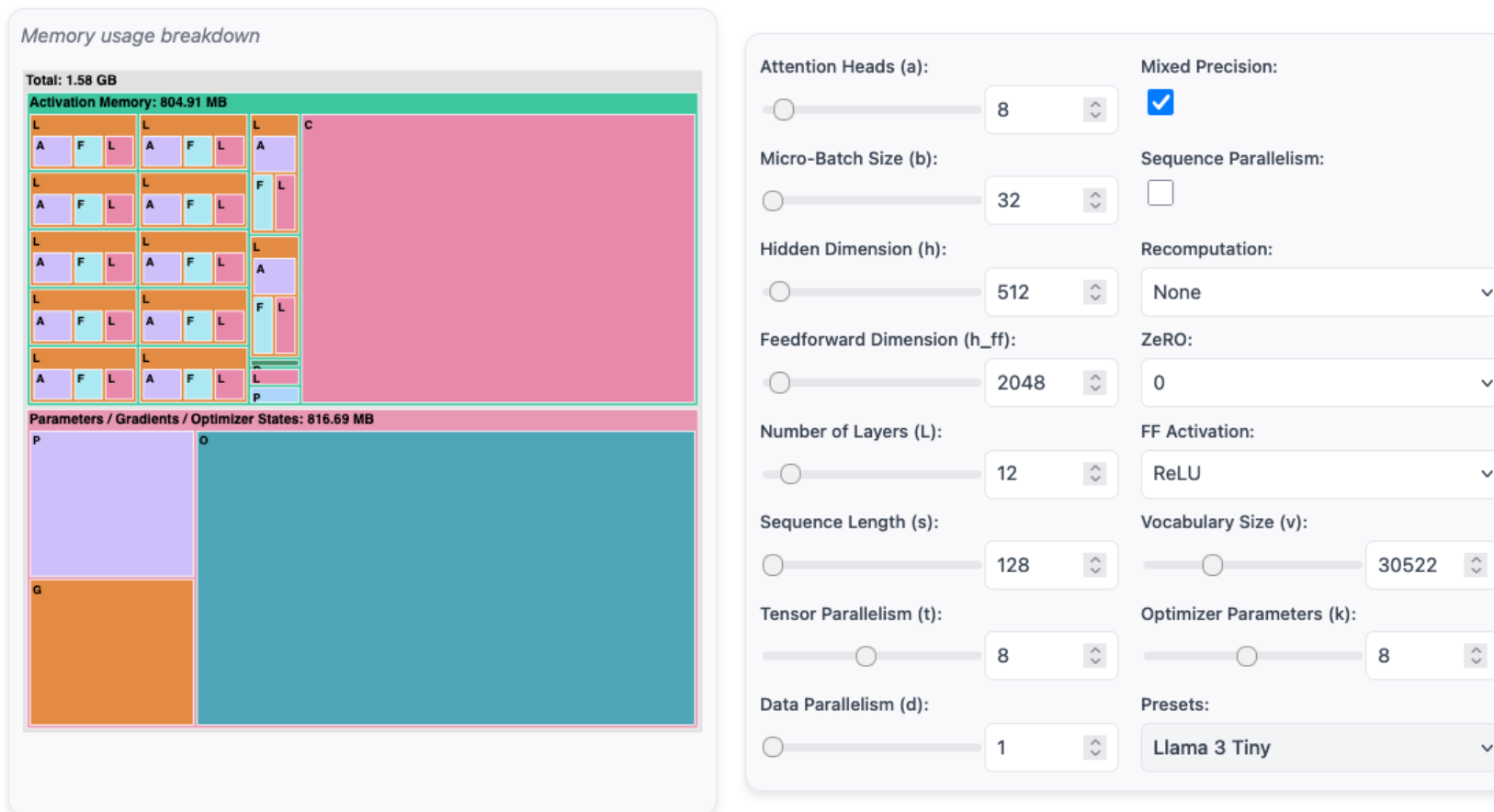
- Partitions optimizer states, gradients, and parameters across data parallel ranks to eliminate redundancy.



Training Frameworks

Framework	Feature Set	Model Support	Unique Features	Community Size	Difficulty
Megatron-LM	Extensive	Kimi-K2, Nemotron	Pioneers of 3D parallelism	93k / 269k	Hard for beginners
DeepSpeed	Extensive	BLOOM, GLM	Pioneers of ZeRO & 3D parallelism	94k / 194k	Hard for beginners
TorchTitan	Growing feature set	PyTorch	Optimised for dense models, MoE improvements underway	7k / 9k	Moderate: requires parallelism know-how
Nanotron	Minimal, tailored for HF pretraining	Yes (StarCoder, SmolLM)	Optimised (UltraScale Playbook)	Not listed	Low: beginner-friendly

Memory Usage



<https://huggingface.co/spaces/nanotron/ultrascale-playbook>